Rossmann Sales Data Wrangling

2026-04-07

Data Wrangling

```
library(tidyverse)
library(lubridate)
```

First, let's load the data

```
train <- read_csv("../data/raw/train.csv")
store <- read_csv("../data/raw/store.csv")
test  <- read_csv("../data/raw/test.csv")

head(train)
```

```
## # A tibble: 6 x 9
##   Store DayOfWeek Date       Sales Customers  Open Promo StateHoliday
##   <dbl>     <dbl> <date>     <dbl>     <dbl> <dbl> <dbl> <chr>
## 1     1         5 2015-07-31  5263       555     1     1 0
## 2     2         5 2015-07-31  6064       625     1     1 0
## 3     3         5 2015-07-31  8314       821     1     1 0
## 4     4         5 2015-07-31 13995      1498     1     1 0
## 5     5         5 2015-07-31  4822       559     1     1 0
## 6     6         5 2015-07-31  5651       589     1     1 0
## # i 1 more variable: SchoolHoliday <dbl>
```

```
head(store)
```

```
## # A tibble: 6 x 10
##   Store StoreType Assortment CompetitionDistance CompetitionOpenSinceMonth
##   <dbl> <chr>     <chr>                    <dbl>                     <dbl>
## 1     1 c         a                         1270                         9
## 2     2 a         a                          570                        11
## 3     3 a         a                        14130                        12
## 4     4 c         c                          620                         9
## 5     5 a         a                        29910                         4
## 6     6 a         a                          310                        12
## # i 5 more variables: CompetitionOpenSinceYear <dbl>, Promo2 <dbl>,
## #   Promo2SinceWeek <dbl>, Promo2SinceYear <dbl>, PromoInterval <chr>
```

We focus on store model a, and aggregate this to form a time series

```
# Join train.csv and store.csv
df_A <- train %>%
  left_join(store, by = "Store") %>%
  filter(StoreType == "a", Open == 1) %>%
  mutate(Date = as.Date(Date))
```

```r
# ==========================
# VERSION 1: CLEAN (UNIVARIATE TS)
# ==========================
df_A_daily_clean <- df_A %>%
  group_by(Date) %>%
  summarise(
    Sales = sum(Sales),
    .groups = "drop"
  ) %>%
  arrange(Date) %>%
  mutate(
    Year = year(Date),
    Month = month(Date),
    Week = week(Date)
  )

head(df_A_daily_clean)
```

```
## # A tibble: 6 x 5
##   Date         Sales  Year Month  Week
##   <date>       <dbl> <dbl> <dbl> <dbl>
## 1 2013-01-01    2907  2013     1     1
## 2 2013-01-02 3920386  2013     1     1
## 3 2013-01-03 3528027  2013     1     1
## 4 2013-01-04 3610215  2013     1     1
## 5 2013-01-05 2964587  2013     1     1
## 6 2013-01-06    6254  2013     1     1
```

```r
# ==========================
# VERSION 2: WITH EXTERNAL VARIABLES
# ==========================
df_A_daily_full <- df_A %>%
  group_by(Date) %>%
  summarise(
    Sales = sum(Sales),
    n_stores = n(),
    Promo = mean(Promo),
    SchoolHoliday = mean(SchoolHoliday),
    StateHoliday = mean(StateHoliday != "0"),
    .groups = "drop"
  ) %>%
  arrange(Date) %>%
  mutate(
    Year = year(Date),
    Month = month(Date),
    Week = week(Date)
  )

head(df_A_daily_full)
```

```
## # A tibble: 6 x 9
##   Date         Sales n_stores Promo SchoolHoliday StateHoliday  Year Month  Week
##   <date>       <dbl>    <int> <dbl>         <dbl>        <dbl> <dbl> <dbl> <dbl>
## 1 2013-01-01    2907        1     0         1                 1  2013     1     1
```

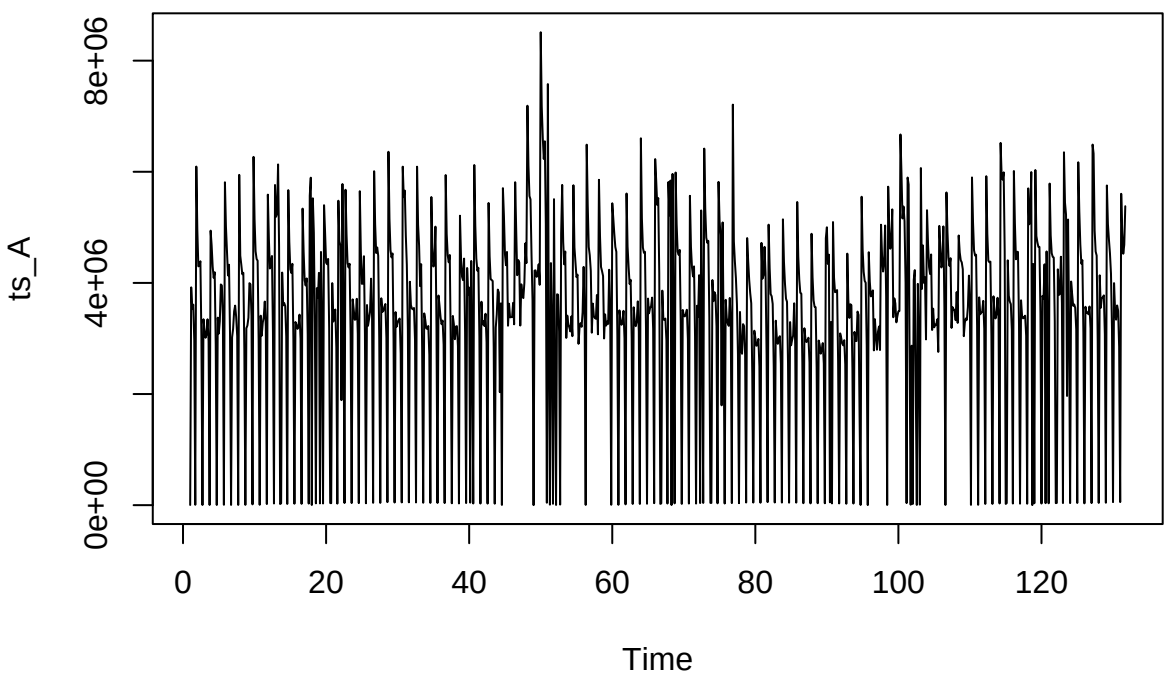
```
## 2 2013-01-02 3920386      602     0         1                0  2013     1     1
## 3 2013-01-03 3528027      601     0         0.938            0  2013     1     1
## 4 2013-01-04 3610215      601     0         0.938            0  2013     1     1
## 5 2013-01-05 2964587      600     0         0.065            0  2013     1     1
## 6 2013-01-06    6254        3     0         0                0  2013     1     1
```

```r
# ==========================
# SAVE BOTH DATASETS
# ==========================
write_csv(df_A_daily_clean, "../data/processed/daily_store_a_clean.csv")
write_csv(df_A_daily_full, "../data/processed/daily_store_a_full.csv")
```

In addition to sales, several time-varying explanatory variables were included. The Promo variable represents the proportion of stores running promotions on a given day, while SchoolHoliday and StateHoliday capture the proportion of stores affected by school and public holidays, respectively. These variables provide additional context for changes in sales beyond purely temporal patterns.

Time series object

```r
ts_A <- ts(df_A_daily_clean$Sales, frequency = 7)
plot(ts_A)
```



The aggregated time series exhibits strong weekly seasonality, with noticeable drops in sales on certain days (e.g., Sundays), reflecting reduced store activity rather than true demand fluctuations. Notice that these drops in sales are due to most of the closed stores on certain days, but it is not entirely zero as some stores remain open that day.